

Renke Wang

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Education

George Mason University <i>Ph.D. Candidate in Electrical and Computer Engineering</i>	<i>August 2022 – Present</i>
Dalian University of Technology <i>Master of Engineering, Control Science and Engineering</i>	<i>Sept 2017 – June 2020</i>
Dalian University of Technology <i>Bachelor of Engineering, Automation</i>	<i>Sept 2013 – June 2017</i>

Professional Summary

Ph.D. candidate in Electrical and Computer Engineering with research experience in real-time scheduling, resource allocation, and control co-design for cyber-physical systems (CPSs) and networked control systems (NCSs). Skilled in developing model predictive control (MPC) frameworks, optimization formulations, simulation pipelines, and visualization tools in Python and MATLAB, with applications to traffic networks, warehouse robots, human-robot collaboration, and robust control.

Research Focus and Technical Expertise

- **Cyber-Physical Systems and Networked Control Systems:** Modeling and control of cyber-physical systems (CPSs), including networked control systems (NCSs), with recurring tasks, timing constraints, and limited shared resources. Applications include traffic networks, human-robot collaboration, warehouse robot coordination, and blimp motion control tasks.
- **Scheduling, Resource Allocation, and Control Co-Design:** Real-time scheduling, resource allocation, and control co-design for multiple dynamical systems under shared-resource constraints, aiming to optimize system performance and resolve resource contention.
- **Centralized and Distributed Optimization Methods:** Nonconvex optimization and mixed-integer programming (MIP) formulations for scheduling problems, addressed using centralized contention-resolving MPC (CR-MPC) and a distributed framework based on the consensus alternating direction method of multipliers (C-ADMM).
- **Model Predictive Control (MPC):** MPC for real-time systems, online scheduling, priority-based scheduling decisions, and constrained control under timing and resource constraints.
- **Robust Control:** Robust control of dynamical systems under timing uncertainties, including scheduling-induced uncertainties, using tube-based MPC.
- **Planning and Search Algorithms:** Tree-based and graph-search methods for optimal scheduling, using DFS, BFS, Dijkstra's algorithm, and A* search, with node evaluation via MPC for multi-step lookahead.
- **Python and MATLAB Implementation:** Algorithm implementation, simulation, parameter sensitivity studies, and solver-efficient local updates for scheduling, control, and optimization problems.

Research Experience

Research Assistant at George Mason University	<i>Fairfax, VA</i>
◦ Distributed Scheduling and Control Co-Design via C-ADMM – Developed a distributed scheduling and control co-design framework based on C-ADMM, with contention-resolving MPC solving local subproblems for preemptive and non-preemptive tasks in NCSs, warehouse robot scheduling, and traffic-network scenarios under shared-resource constraints. ◦ Centralized Scheduling and Control Co-Design via Contention-Resolving MPC – Developed contention-resolving MPC for scheduling and control co-design under heterogeneous resource constraints, with applications to human-robot collaboration and game-theoretic formulations for multi-robot obstacle avoidance. – Formulated and applied contention-resolving MPC to systems with determined resource occupation orders, including multi-intersection traffic networks and multi-lane intersection scheduling.	

- Robust MPC for NCSs under Distribution-Free Timing Perturbations
 - Developed a tube-based robust MPC framework for NCSs under distribution-free timing perturbations, designed dynamic controller gains, and established stability guarantees for closed-loop error dynamics.
- Visualization of Scheduling Results for Warehouse and Traffic Scenarios
 - Built browser-based visualization and simulation workflows for warehouse robot and traffic-network scheduling scenarios, including vehicle/robot motion, resource-occupation timelines, scheduling plots, cost evolution, and comparisons between optimized schedules and FCFS baselines.

Teaching Assistant at George Mason University

Fairfax, VA

- Continuous-Time Signals and Systems (Laboratory course): Teaching and co-teaching lab sessions; grading assignments; providing academic support for undergraduate students.
- Classical Systems and Control Theory: Grading coursework and providing class support to assist instruction and exam preparation.

Master Student at Dalian University of Technology

Liaoning, Dalian

- Conducted research on stability analysis and control design of positive and switched systems, focusing on fuzzy control, Markov jump systems, and robust stabilization methods.

Journal Articles and Manuscripts

- Renke Wang and Ningshi Yao. *Distributed Scheduling and Control Co-Design for Networked Control Systems with Multi-Layered Shared Resources*. To be submitted to IEEE Transactions on Control of Network Systems (TCNS).
- Renke Wang, Mengxue Hou, and Ningshi Yao. *Robust Co-Design of Scheduling and Control for Lighter-than-Air Agents with Heterogeneous Resources*. International Journal of Robust and Nonlinear Control, under review.
- J. Lian and R. Wang. *Stochastic Stability of Positive Markov Jump Linear Systems with Fixed Dwell Time*. Nonlinear Analysis: Hybrid Systems, 40 (2021): 101014.

Conference Papers

- R. Wang and N. Yao. *Contention-Resolving Model Predictive Control for Coordinating Automated Vehicles at a Multi-Lane Traffic Intersection*. 2025 IEEE Conference on Control Technology and Applications (CCTA). IEEE, 2025.
- Rajul Kumar, Renke Wang, and Ningshi Yao. *One Human and Multi-Robot Collaboration: Evaluating the Impact of Attention-Guided On-Screen Recommendations on Human Multitasking Efficiency, Cognitive Load, and Trust*. 2025 34th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), IEEE, 2025.
- R. Wang and N. Yao. *Robust Model Predictive Control for Networked Control Systems with Timing Perturbations*. 2024 American Control Conference (ACC). IEEE, 2024.
- N. Yao and R. Wang. *Optimal Real-Time Human Attention Allocation and Scheduling in a Multi-human and Multi-robot Collaborative System*. 2024 IEEE Conference on Control Technology and Applications (CCTA). IEEE, 2024.
- J. Lian, R. Wang, and F. Wu. *Exponential Stability Analysis and Controller Design for LTI Positive System with Controller Failure*. In 12th Asian Control Conference, Kitakyushu, Japan, 2019, pp. 179-184.

Technical Skills

Python

- Implemented Python versions of scheduling, control, and optimization algorithms for C-ADMM, contention-resolving MPC, and multi-agent scheduling under shared-resource constraints.
- Implemented tree-based and graph-search algorithms for scheduling and planning, including DFS, BFS, Dijkstra's algorithm, and A* search with multi-step lookahead evaluation.
- Developed numerical optimization and control implementations, including MIP and LMI-based formulations, online MPC prototypes using CasADi, and local optimization updates for distributed scheduling problems.

- Built simulation and data-processing workflows for traffic-network and warehouse robot scheduling, including scheduling plots, cost evolution, and performance evaluation.

MATLAB

- Developed original research prototypes for scheduling, control, and optimization algorithms, including C-ADMM, contention-resolving MPC, robust MPC, and timing-perturbation analysis.
- Built early constrained optimization models using YALMIP/fmincon and related MATLAB toolboxes for MIP, LMI, and MPC-based control subproblems.
- Conducted simulation-based performance evaluation for scheduling and control co-design frameworks.

Tools

- Git/GitHub, LaTeX, CasADi, NumPy/SciPy, Matplotlib, YALMIP/fmincon.

Honors and Awards

- Summer Research Assistantship Award, George Mason University Graduate Division, 2026

Professional Service

Reviewer for IEEE technical conferences and journals, 2023–2026.